CSWP: march on in time

Alichan Borokov
Alichan.Borokov@UGent.be

Sam Hemati
Sam.Hemati@UGent.be

Joran Van Haelst
Joran.VanHaelst@UGent.be

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1 Summary of the problem

In this assignment we implement the march-on in time algorithm on PEC scatterers. We implemented on TM case of the wave equation. This means that the electric field only contains a z-component and the magnetic field only in the x and y- direction. the problem is also consistent in the z-direction this means the material is symmetric under the z-axis. This reduces our problem to a 2 dimensional problem.

After that we use the greens function in its retarded form to solve the boundary integral equation which has as the unknown the current. At last we discretize the integral equation as done in the assignment. And we us those formulae to describe the different problems.

2 assignment

we validate our code by calculating the time domain scattering at a cylinder.

The formula description is done in the assignment. Our task is implement the march-on-in-time algorithm on this example and look if this correspondds with the analytical solution.

for the first part we calculate the timedomain scattering at a cylinder. Using an incoming wave going to the positive x direction.

3 Implementation

The algorithm is implemented in Python, using the Numpy and Scipy packages.

The MOT algorithm is fully contained in the `MOT` class, initiating an instance takes a few arguments:

param	value	unit	description
R	10	m	radius of PEC
t_0	9e-8	s	center of incident pulse
T	20	m	width of incident pulse
dt	1e-9	s	timestep
t_{end}	64e-8	s	end of simulation
N_S	32	1	number of segments for PEC
N_G	8	1	order of Gaussian quadrature

Table 1: Parameters for validation

- `curve` is a `numpy` array of shape $(2, N_S + 1)$. The first index is the coordinate (`0` for x or `1` for y, generally `_x` and `_y` were written as aliases for 0 and 1). N_S is the number of segments in which the curve has been discretized. This field consists out of the edges of these segments (the first and last elements must equal),
- `dt` is the time step in seconds,
- `N_T` is the number of timesteps taken, and
- `N_G` is the number of points for the Gaussian quadrature.

With these arguments the quadrature points (`coordquad` and `weights`), the geometry fields (`rho` are the midpoints, `tangents` the tangent vectors and `l` the lengths of the segments, `rhop[_x, m, g]` is the x-coordinate of the m-th segment, linearly interpolated for x_g), F (an array of shape (N_T, N_S, N_S, N_G), such that $F(k, \rho_m, \rho_{n,g})$ = `F[k,m,n,g]`) and Z (with shape (N_T, N_S, N_S), so that $(Z_k)_{m,n}$ = `Z[k,m,n]`) are calculated.

Finally the function `solve` implements the stepping algorithm. It takes one argument `V`, which is an array of shape (N_T, N_S), such that $(V_j)_m = E^i(\rho_m, j\Delta t)$ = `V[j,m]`. The solver (and `_init_Z`) uses the Numpy function `einsum`, which makes translating sums from math into code a lot easier:

$$\sum_{g}^{N_G} w_g F(k, \rho_m, \rho_{n,g}) = \texttt{np.einsum('g,kmng->kmn', weights, F)}$$

This and a few reshapes which had to happen for Numpy to be able to correctly broadcast together some arrays, are the only noteworthy difference between the code and its mathematic description from the assignment.

TODO maybe some stuff about the second excercise and the E^i calculation.

4 Validation

For the validation the MOT-algorithm is ran with as incident field a pulse. The incident wave is plotted in figure ?? at the 'sun'- and shadow-side. The parameters T and t_0 are chosen in such a manner that the incident field is zero at $t = 0$ over the whole PEC and such that the width in the frequency domain is wide enough to cover the range $\omega/c \in [0, 1]$. Their values are given in table **??**.

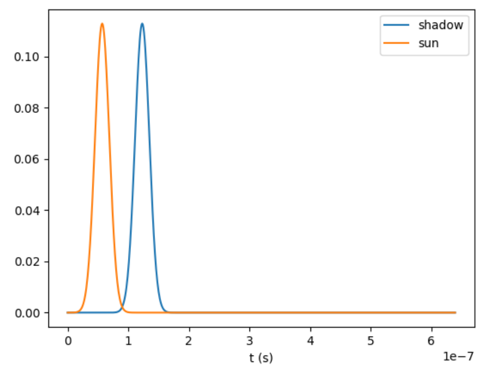

Figure 1: Incident field at the sun- and shadow-side

The solution of the MOT-algorithm can easily be integrated to obtain the current induced on the PEC. It is shown in figure **??** in the form of a polar plot and in figure **??** as a function of time.

The spectrum of the incoming wave quickly decays to zero. This means that when dividing by this spectrum the result will explode. Due to this the frequency range for which the solver can be used is only limited. We chose to limit the frequency range to the region where the spectrum of the incoming wave is more than 0.1 % of its maximum.

When calculating the Fourier transform of the solution from the MOT-algorithm, Numpy's `rfft` function was used. It is important to note that this is a discrete transform without any normalization factor. To compare it with the continuous transform, a correction factor of $dt/\sqrt{2\pi}$ is needed.

4.1 Analytical Solution

$$e_z(\vec{\rho}) = \sum_{n=-\infty}^{\infty} j^n \left(J_n(k\rho) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) \right) e^{jn\phi} \tag{1}$$

The analytical solution (**??**) for e_z is copied from appendix B from the assignment. The derivate towards ρ is taken and evaluated at $\rho = a$:

$$\sum_{n=-\infty}^{\infty} j^n \left(kJ_n'(ka) - k\frac{J_n(ka)}{H_n^{(2)}(ka)} H_n'^{(2)}(ka) \right) e^{jn\phi}.$$

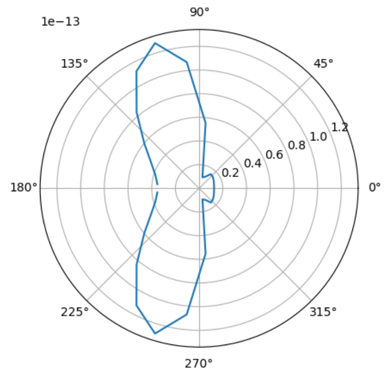

Figure 2: Polar plot of the current at $t = t_0$

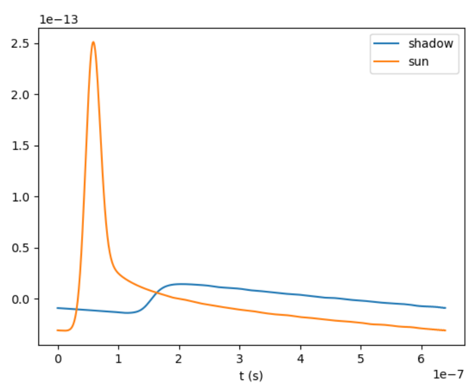

Figure 3: Plot of the current at the sun- and shadow-side

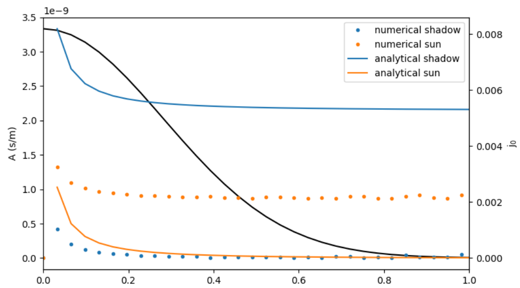

Figure 4: Current at the sun-side and shadow-side calculated numerically and analytically

With a bit of refactoring,

$$\sum_{n=-\infty}^{\infty} \frac{j^n k}{H_n^{(2)}(ka)} \left(J_n'(ka) H_n^{(2)}(ka) - J_n(ka) H_n'^{(2)}(ka) \right) e^{jn\phi},$$

the Wronskian relationship is revealed, so that finally equation **??** follows.

$$j_z = \frac{1}{j\omega\mu} \sum_{n=-\infty}^{\infty} \frac{j^n k}{H_n^{(2)}(ka)} \frac{2j}{\pi ka} e^{jn\phi} \tag{2}$$

The summation in this equation was cut off at $ka + 2 \approx N \in Z$, which was established by the rule of thumb that $H_n^{(2)}(z)$ decays exponentially as soon as $n > z$ in combination with some experimentation.

4.2 Discussion

In figure **??** the numerically and analytically computed current in the frequency domain are plotted, alongside with the spectrum of the incoming wave (in black). The analytical and numerical current follow the same shape, but don't match at all. Even stranger is that the shadow-side from the analytical solution seems to lie higher than its sun-side counterpart. ...

The width of the incoming pulse can be made smaller, so that in the frequency domain it becomes wider. This cannot be continued indefinitely, as the timestep dt becomes too small to represent the pulse correctly in the timedomain. Making dt smaller makes that the pulse more correctly approximates a Gaussian pulse, but also makes the maximum frequency (after the Fourier-transform) bigger. At these maximum frequencies the spectrum of the incoming pulse is again nearly zero, so that the additional range in frequencies gained is useless. Thus a tradeoff between the width of the pulse and the timestep must be made. Another parameter that can be adjusted is t_{end}. As this increases the resolution in the frequency domain also increases, so that finer details are visible. Increasing this (or decreasing dt) comes at a computational cost, as these increase the total steps in time N_T, so that the computation takes longer. Increasing the radius of the PEC increases the difference between the current at the shadow-side and at the 'sun'-side. The same effect can also be accomplished by decreasing the width of the incoming pulse.

Lastly N_S and N_G can be changed. The number of segments must be great enough so that the geometry of the PEC is correctly approximated. Increasing this any further does not increase the accuracy very much. Increasing N_S also comes at the cost of requiring more memory. The order of the Gaussian quadrature does not have much influence. As, once the geometry is well approximated, ρ_m are close toghether, so that $|\rho_m - \rho'|$ in F is small and can easily be approximated by a low order polynomial.

4.3 Reproduction

All plots (along with some extra animations) in the validation section can be reproduced by executing the function `Q_6_1_validation` near the end of the file, with as arguments the parameters given by table **??**.

5 Resonance Frequencies of Cylindrical Cavity

for the cylindrical cavity we start with the current source

$$J(\boldsymbol{\rho}, t) = \frac{4}{T\sqrt{\pi}} e^{-(4c\frac{t-t_o}{T})^2} \frac{\delta(\rho)}{2\pi\rho} \tag{3}$$

5.1 Incident Field by Numerical Quadrature

To compute the incident field, we follow the assignment instructions and use the two–dimensional time–domain Green's function. The resulting expression for the incident electric field is evaluated numerically via temporal quadrature.

We begin with the time–domain integral equation (cf. Eq. (19)), which gives the z–directed electric field generated by a volumetric source distribution,

$$E_z(\boldsymbol{\rho}, t) = -\frac{\mu}{2\pi} \int \frac{H!\,(t - |\boldsymbol{\rho} - \boldsymbol{\rho}'|/c)}{\sqrt{(t-\tau)^2 - |\boldsymbol{\rho} - \boldsymbol{\rho}'|^2/c^2}}$$

$\partial_t J_z(\boldsymbol{\rho}', t), dV', (4)$ where $H(\cdot)$ is the Heaviside step function and $*$ denotes convolution in time.

Point source model. The source is assumed to be located at the origin. Using polar coordinates $\boldsymbol{\rho}' = (\rho', \phi)$ and a Dirac delta distribution to represent the point source, the current density is written as

$$J_z(\boldsymbol{\rho}', t) = \frac{4}{T\sqrt{\pi}} \exp! \left[-\left(\frac{4c(t-t_0)}{T} \right)^2 \right] \frac{\delta(\rho')}{2\pi\rho'}. \tag{5}$$

Substituting this expression into the integral and carrying out the spatial integration yields

$$E_z(\rho,t) = -\frac{\mu}{2\pi}\frac{H(t-\rho/c)}{\sqrt{t^2-\rho^2/c^2}}$$

$\partial_t\left(\frac{4}{T\sqrt{\pi}}\exp!\left[-\left(\frac{4c(t-t_0)}{T}\right)^2\right]\right)$, (6) where $\rho = |\boldsymbol{\rho}|$.

Temporal convolution. Evaluating the convolution explicitly gives

$$E_z(\rho,t) = -\frac{\mu}{2\pi}\int_{-\infty}^{t}\frac{H(t-t'-\rho/c)}{\sqrt{(t-t')^2-\rho^2/c^2}},\partial_{t'}\left(\frac{4}{T\sqrt{\pi}}\exp!\left[-\left(\frac{4c(t'-t_0)}{T}\right)^2\right]\right)dt'. \tag{7}$$

The time derivative of the source pulse is

$$\partial_{t'}\left(\frac{4}{T\sqrt{\pi}}\exp!\left[-\left(\frac{4c(t'-t_0)}{T}\right)^2\right]\right) = -\frac{64c^2}{T^3\sqrt{\pi}}(t'-t_0)\exp!\left[-\left(\frac{4c(t'-t_0)}{T}\right)^2\right]. \tag{8}$$

The Heaviside function restricts the domain of integration to

$$t' \le t-\rho/c, \tag{9}$$

so that the convolution reduces to

$$E_z(\rho,t) = \frac{32\mu c^2}{\pi T^3\sqrt{\pi}}\int_{-\infty}^{t-\rho/c}\frac{(t'-t_0)\exp!\left[-\left(\frac{4c(t'-t_0)}{T}\right)^2\right]}{\sqrt{(t-t')^2-\rho^2/c^2}},dt'. \tag{10}$$

Change of variables. For numerical evaluation it is convenient to introduce the retarded–time variable

$$\tau = t-t', \qquad dt' = -d\tau. \tag{11}$$

With this substitution, the integration limits become $\tau \in [\rho/c,\infty)$, and the incident field takes the form

$$E_z(\rho,t) = \frac{32\mu c^2}{\pi T^3\sqrt{\pi}}\int_{\rho/c}^{\infty}\frac{(t-\tau-t_0)\exp!\left[-\left(\frac{4c(t-\tau-t_0)}{T}\right)^2\right]}{\sqrt{\tau^2-\rho^2/c^2}},d\tau. \tag{12}$$

Cosh substitution. The square–root singularity in the denominator can be removed by introducing the hyperbolic substitution

$$\tau = \frac{\rho}{c}\cosh u, \qquad d\tau = \frac{\rho}{c}\sinh u, du, \tag{13}$$

for which

$$\sqrt{\tau^2-\rho^2/c^2} = \frac{\rho}{c}\sinh u. \tag{14}$$

Under this change of variables, the lower limit $\tau = \rho/c$ corresponds to $u = 0$, and the upper limit $\tau \to \infty$ corresponds to $u \to \infty$. The integral therefore simplifies to

$$E_z(\rho, t) = \frac{32\mu c^2}{\pi T^3 \sqrt{\pi}} \int_0^{\infty} \left(t - t_0 - \tfrac{\rho}{c} \cosh u\right) \exp! \left[- \left(\frac{4c}{T} \left(t - t_0 - \tfrac{\rho}{c} \cosh u\right) \right)^2 \right] , du. \tag{15}$$

This cosh substitution eliminates the square–root singularity and yields an integrand that is smooth and rapidly decaying, making it particularly suitable for stable and accurate numerical quadrature. This final expression is used to compute the incident field in the time domain.

Finite interval reduction and Gauss–Legendre mapping. Causality imposes an upper bound on the hyperbolic variable u. Since the source pulse is effectively supported only for arguments $t - \tau - t_0 \leq 0$, the integrand vanishes once

$$\tau = \frac{\rho}{c} \cosh u > t - t_0. \tag{16}$$

This yields the finite upper limit

$$u_{\max} = \operatorname{arccosh} \left(\frac{c}{\rho} (t - t_0) \right), \tag{17}$$

and the incident field becomes

$$E_z(\rho, t) = \frac{32\mu c^2}{\pi T^3 \sqrt{\pi}} \int_0^{u_{\max}} \left(t - t_0 - \tfrac{\rho}{c} \cosh u\right) \exp \left[- \left(\frac{4c}{T} \left(t - t_0 - \tfrac{\rho}{c} \cosh u\right) \right)^2 \right] du. \tag{18}$$

To apply Gauss–Legendre quadrature, the finite interval $u \in [0, u_{\max}]$ is mapped to the standard interval $\xi \in [-1, 1]$ using the linear transformation

$$u = \frac{u_{\max}}{2} (1 + \xi), \qquad du = \frac{u_{\max}}{2} \, d\xi. \tag{19}$$

Substitution yields

$$E_z(\rho, t) = \frac{16\mu c^2 u_{\max}}{\pi T^3 \sqrt{\pi}} \int_{-1}^{1} \left(t - t_0 - \tfrac{\rho}{c} \cosh u(\xi)\right) \exp \left[- \left(\frac{4c}{T} \left(t - t_0 - \tfrac{\rho}{c} \cosh u(\xi)\right) \right)^2 \right] d\xi, \tag{20}$$

where $u(\xi) = \frac{u_{\max}}{2}(1 + \xi)$. This integral is evaluated numerically using Gauss–Legendre quadrature. This is done in the code at Ei2 code and runs smoothly

5.2 Resonance Modes

to calculate the resonance modes in our cavity we start to calculate the sourceless resonance mode from the sourseless wave equation in the frequency domain. This is

$$\nabla_t^2 e_z(\rho, \phi) + k^2 e_z(\rho, \phi) = 0 \tag{21}$$

in the 2 dimensional case the solution are the besselfunction in the form like

$$R(\rho) = AJ_n(k\rho) \tag{22}$$

the Bessely function dissapears because it is non-physical at $rho = 0$ in the ent you have

$$e_z = R(\rho)\Phi(\phi) = \sum_{n=-\infty}^{\infty} A_n J_n(k\rho)e^{jn\phi}$$

the zeroes of this function is when $\rho = a$ and thus we find the it that $J_n(x) = 0$ with $x = ka$ with $k = \omega/c)$ thus you have $J_n(\omega/c \cdot a)$ you have $\omega_{m,n} = \frac{c}{a}x_{n,m}$

which stands for the mth zero and the nth order bessel function.

5.3 Spectrum Numerical Solution

We are looking at the spectrum of the current and compare with the current coming from the analytical solution. This is done in the validation 62 code.

5.4 Off-center Source

6 Creative Assignment